

Is the RAG Action Pool Routable? A Failure-First Study of Grounding Utility and GPU Energy

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Abstract—RAG routers are usually optimized as if their candidate actions already contain a useful, query-dependent trade-off. That premise is testable before a learned policy is credited. Using sealed traces, we retrospectively apply an integrated failure-first audit that separates the best fixed action from a label-aware per-query oracle. Their difference, routing headroom, quantifies whether selection can improve strict utility at all. The controlled study contains 800 grouped scenarios, eight evidence and memory tasks, five pinned generators, and 17 retained direct and composite-grounding actions. The sealed test contains 2,040 route-query traces. Success jointly requires answer correctness, exact required-evidence citations, perfect citation precision, correct abstention, parseability, and no forbidden evidence. We also measure selected-T4 board energy during generation; retrieval and whole-system energy lie outside that boundary.

The 11-action resident-eligible pool had a best fixed success of 12.5% and an oracle ceiling of 12.5%: all 15 successes came from one direct route, leaving 0.0 percentage points of routing headroom. By contrast, matched direct/grounded pairs for three 3B–4B references had 13.3–21.7 points of headroom, and the complete 17-action matrix had 29.2 points. Retrieval was identical across grounded generators and complete for 80.0% of evidence-required questions, yet strict success ranged from 0% to 58.9%, localizing much of the variation after evidence acquisition. A task-aware, non-learned direct/grounded oracle improved both success and measured generation energy relative to always grounding for the capable references. The frozen learned controller nevertheless had no validation-feasible threshold, selected one fail-closed fallback for every test query, achieved 0%, and saved no energy. The study therefore identifies action-pool routability as an empirical prerequisite: calibration cannot manufacture capability or complementarity absent from the eligible actions.

Index Terms—retrieval-augmented generation, language-model routing, action-pool feasibility, GPU energy, grounding, abstention, negative results

I. Introduction

Learned routers are judged by the actions they select, but their attainable value is bounded before learning begins. If every candidate fails on the same queries, a perfect selector cannot improve utility over the best fixed action. If the only successful action is already the least expensive, cost-aware selection has no remaining trade-off to exploit. A router trained in either regime may be calibrated and computationally efficient while solving an empty decision problem. We call the missing prerequisite action-pool routability: the pool must contain adequate capability, query-dependent complementarity, measurable costs, and a feasible rule for selecting among them.

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This prerequisite is easy to overlook in retrieval-augmented generation (RAG). RAG is not a single switch. A composite action can change the generator, sparse or dense retrieval, second-hop expansion, external-memory organization, prompt length, citation format, deterministic verification, and abstention policy. Recent systems route among retrieval-augmented generators, retrievers, or direct/RAG/reasoning pipelines [1]–[4]. These methods answer how to choose among useful options. They do not remove the need to establish that the eligible options are useful and complementary in the first place.

The issue is particularly acute for compact generators. Retrieved evidence helps only when a model can attend to it, produce the requested schema, cite the exact support, and abstain when support is absent. Model, task, context, and retrieval characteristics jointly affect this behavior [5]; imperfect or distracting context can reduce answer quality [6], [7]. Grounding also adds work. Existing studies profile accuracy against tokens, latency, monetary cost, or energy [8]–[11]. A scientific audit must therefore state both its utility contract and its measurement boundary rather than treating “quality” or “energy” as interchangeable across systems.

We retrospectively apply an integrated failure-first audit to a controlled diagnostic study of small-model RAG and memory actions. Adapting the established single-best/virtual-best comparison from algorithm selection, the key diagnostic compares the best fixed strict-success rate with a label-aware per-query oracle over the same action pool. Their difference is routing headroom. A zero value certifies that no selector can improve strict success on that matrix, regardless of model architecture or calibration. This oracle is deliberately post hoc: it diagnoses the opportunity present in sealed traces and is not reported as a deployable policy [12]–[14].

The empirical testbed contains 800 generated scenario groups and eight task classes covering direct copying, document evidence, personal memory, composition, temporal updates, authority conflicts, two-hop evidence, and deleted or missing facts. Five pinned FP16 generators are compared through matched direct and composite-grounding endpoints. Strict success requires answer correctness, exact support, valid abstention, parseability, and exclusion of forbidden evidence. Physical telemetry is restricted to selected-board GPU energy during generation; it is not a measure of total RAG energy or carbon.

The study asks four questions:

RQ1 Does the resident-eligible action pool contain adequate capability and query-dependent routing head-

room?

- RQ2 When retrieved evidence is held constant, where does the pipeline lose utility between retrieval, generation, citation, and verification?
- RQ3 How do generator and task class alter the success and GPU-generation-energy effect of the composite grounding intervention?
- RQ4 Can the frozen learned controller satisfy its validation constraints, and does its sealed-test behavior agree with the pool audit?

Across 11 resident-eligible actions, the best fixed route and the per-query oracle both succeeded on 15 of 120 questions (12.5%); every success belonged to the same direct action. In contrast, the six offline reference actions had 28.3 percentage points of headroom, and the full 17-route matrix had 29.2 percentage points. Thus, complementarity existed in the experiment but was excluded from the learned controller’s action set. The remaining analyses explain that separation. All five grounded endpoints received identical retrieved evidence, yet their evidence-task success ranged from 0% to 58.9%. For the capable reference generators, choosing direct operation on the two evidence-free task classes and grounding on the other six improved strict success and used less measured generation energy than always grounding. The learned controller, however, had no validation-feasible threshold and reproduced a zero-success fallback on every test query.

The paper makes three contributions:

- 1) an integrated failure-first protocol that adapts the established single-best/oracle gap to evidence-qualified composite RAG actions and combines it with interface, capability, physical-cost, and constraint-feasibility checks;
- 2) a controlled retrieval-to-utilization analysis showing that identical evidence can yield sharply different schema, citation, answer, and strict-success outcomes across generators and task classes;
- 3) a sealed negative case study that couples strict evidence-qualified utility with a precisely bounded physical GPU-energy measure and preserves validation infeasibility instead of repairing the policy after test access.

The claim is not that a new routing algorithm outperforms prior work. It is that routing opportunity is an empirical property of the eligible action pool and should be audited before optimization. In this retrospective case, the audit provides an impossibility certificate for strict-success improvement within the resident-eligible matrix and identifies where utility complementarity remains.

II. Related Work and Unresolved Gap

A. Adaptive Retrieval and Composite Routing

RAG couples a parametric generator to retrieved non-parametric evidence [15]. Adaptive systems subsequently made retrieval conditional: FLARE retrieves around uncertain continuations [16], Adaptive-RAG selects among retrieval depths [17], and Self-RAG learns retrieval and

critique decisions [18]. Parallel work routes among language models to trade quality against monetary or computational cost [19]–[21].

Our headroom statistic is not a new algorithm-selection construct. Per-instance algorithm selection has long compared a single best solver with a label-aware virtual best solver [12], [13]. In contemporary LLM routing, LLMRouterBench explicitly treats model complementarity as the premise for routing and measures a remaining Oracle gap [14]. We adapt this established comparison to composite RAG actions scored by evidence-qualified utility. The contribution is the integrated five-question audit—interface compatibility, capability, complementarity, cost comparability, and constrained selectability—and its application to immutable traces, not the subtraction of oracle and single-best scores by itself.

Recent methods join these ideas more directly. RAGRouter conditions generator selection on retrieved documents [1]; R^3 AG separates retrieval quality from downstream generation utility when routing retrievers [2]. X-Router chooses among direct, RAG, chain-of-thought, and combined pipelines [3]; RouteRAG selects text or graph evidence and reasoning actions [22]; and RAG-on-a-Diet makes model choices at individual reasoning hops [4]. These works establish retrieval-aware and composite-action routing as active research areas. Accordingly, our novelty is not the existence of a RAG router or an oracle gap. We ask whether the established complementarity diagnostic, combined with interface, physical-cost, and validation-feasibility checks, would have rejected the action pool before policy optimization.

B. Context Utility and Evidence-Qualified Evaluation

The value of retrieved context depends on model capability, task, context length, and retrieval properties [5]. Irrelevant, misleading, or internally conflicting context can impair generation [6], [7], while ASTUTE RAG explicitly targets imperfect retrieval and parametric-context conflicts [23]. Robustness methods such as RoseRAG and PRA-RAG address noise or retrieval corruption with learned and provable defenses [24], [25]. Our four corruptions are therefore a diagnostic negative control, not a new defense or a broad adversarial-robustness claim.

Answer accuracy alone can hide unsupported generation. KILT links answers to provenance [26]; ALCE evaluates citation correctness and completeness [27]; and GaRAGE provides grounding annotations and deflection-oriented evaluation [28]. We combine answer, exact citation, abstention, parsing, and forbidden-evidence conditions in a single strict indicator so that a route cannot receive utility for an ungrounded correct string. The resulting metric is intentionally operational and harsher than human judgment; it does not establish semantic entailment.

Synthetic testbeds provide experimental control but can exaggerate context-use patterns that do not survive realistic retrieval [29]. We therefore describe HybridBench as a mechanism identification testbed. It isolates failures

TABLE I

Scope relative to representative routing and efficiency studies. “Support-qualified” means that utility explicitly depends on evidence use rather than answer quality alone. “Pre-fit audit” names the components evaluated before fitting a selector; LLMRouterBench includes complementarity and an Oracle gap, whereas our five-part audit additionally tests interfaces, absolute capability, physical-cost comparability, and constraint feasibility. This is not a performance ranking.

Study	Routed unit	Reported cost signal	Support-qualified	Physical energy	Pre-fit audit
Adaptive-RAG	no / single / iterative retrieval	retrieval depth	no	no	no
RAGRouter	retrieval-augmented generator	latency	no	no	no
X-Router	direct / RAG / CoT composition	tokens and latency	no	no	no
LLMRouterBench	language-model endpoint	price and latency	no	no	oracle gap
Electro	retriever, depth, and generator configuration	measured energy	no	yes	no
This study	generator, retrieval, memory, and guard action	T4 generation energy	yes	yes	five gates

under known evidence requirements; it does not estimate open-domain QA performance.

C. External Memory

External-memory research ranges from separate long-term memory networks [30] and user-history stores [31] to tiered memory management [32]. More recent systems add hierarchical storage and updating [33], learned memory operations [34], and temporal graph structures with conflict resolution [35]. Our flat, hierarchical, and graph/temporal routes are deliberately simpler. They are controlled action variants used to test whether memory organization creates routable differences under a fixed prompt budget; they are not proposed as new memory architectures.

D. Energy and Measurement Boundaries

Green-AI work argues that computational cost should accompany predictive quality [8], [9], [36], and inference can dominate the operational footprint of deployed models [37]. For RAG specifically, Electro profiles retriever, retrieval-depth, and generator configurations in an accuracy–energy landscape [10]; a controlled software engineering study examines energy-reduction techniques across RAG configurations [11]. These studies preclude a claim that energy-aware RAG configuration is new.

Our distinction is narrower. We pair an explicit evidence-qualified outcome with physical T4 board-power integration during generation, disclose cross-board non-comparability, and use the resulting trace matrix to audit whether the action pool has a selectable quality–cost structure. The contribution is the pre-optimization feasibility diagnosis, not broader energy coverage or a new power instrumentation method.

III. Failure-First Action-Pool Audit

A. Composite Actions and Strict Utility

The single-best/virtual-best gap is established in per-instance algorithm selection [12], [13] and has a direct

LLM-routing analogue in LLMRouterBench’s Oracle analysis [14]. We operationalize that comparison for composite RAG actions, then place it inside a broader audit that also checks interfaces, absolute capability, physical-cost comparability, and validation feasibility.

Let $\mathcal{A}$ be the actions available to a router and let q_i denote one of N evaluation queries. An action fixes the generator, retrieval mechanism, memory organization, prompt contract, and deterministic guard. Its strict outcome is $S_{ia} \in \{0,1\}$. Because the outcome includes answer, evidence, abstention, and interface requirements, two routes that produce the same answer string need not have the same utility.

The best fixed utility of an action pool is

$$B(\mathcal{A}) = \max_{a \in \mathcal{A}} \frac{1}{N} \sum_{i=1}^N S_{ia}. \quad (1)$$

A label-aware oracle that chooses any successful action separately for each query has utility

$$O(\mathcal{A}) = \frac{1}{N} \sum_{i=1}^N \max_{a \in \mathcal{A}} S_{ia}. \quad (2)$$

We refer to the difference as routing headroom:

$$H(\mathcal{A}) = O(\mathcal{A}) - B(\mathcal{A}). \quad (3)$$

O is an upper bound derived from outcomes, not a policy that can be executed without labels. If O is below a required utility $q_{\min}$, the action pool is capability-infeasible for that requirement. If $H = 0$, no selector can improve strict success over the best fixed action on the observed matrix. Positive headroom is necessary but not sufficient: a practical policy must still predict the useful differences from information available before generation and generalize beyond the matrix.

B. Cost Structure

For route a , let E_{ia} be the selected-board generation energy when telemetry is valid. Mean cost is $\bar{E}_a = N^{-1} \sum_i E_{ia}$. An action a descriptively dominates b in the observed quality–energy plane when $\bar{S}_a \geq \bar{S}_b$ and $\bar{E}_a \leq \bar{E}_b$, with at least one strict inequality. Physical interpretation additionally requires a common measurement boundary and attention to device assignment. In this study, all traces use the same generation-window definition, but not every pair ran on the same T4 board; cross-board dominance is therefore descriptive rather than causal.

An energy-aware router can have value even when its success headroom is zero if multiple actions cover the same successful queries at meaningfully different costs. If the best fixed action covers every successful query and also has the lowest route-level mean energy, then no other fixed route improves the observed mean-quality/mean-energy point. This statement does not exclude small per-query cost substitutions caused by trace-level energy variation; we audit that label-aware bound separately rather than equating route diversity with a useful Pareto frontier.

C. Five Audit Questions

The proposed audit proceeds from route validity to policy feasibility:

- 1) Interface compatibility: do retained model–prompt–runtime combinations terminate and produce parseable, support-checkable outputs?
- 2) Capability: does $O(\mathcal{A})$ meet the declared operating requirement, or is the action set inadequate even with label access?
- 3) Complementarity: is $H(\mathcal{A}) > 0$, and which actions add unique successful queries?
- 4) Cost comparability: are costs observed under an explicit common boundary, and do successful actions expose a genuine quality–cost choice?
- 5) Constrained selectability: can a decision-time policy satisfy validation requirements for success, coverage, latency, and abstention?

Failure at an early question changes the interpretation of later stages. An incompatible action should be repaired under a new specification or removed before expensive matrix collection. A capability-infeasible pool should be expanded before router fitting. Zero complementarity distinguishes action-space failure from classification error. A measurement qualification limits energy claims. Finally, validation infeasibility should produce an explicit non-deployment state rather than a threshold chosen after inspecting test labels.

Figure 1 summarizes these questions and the observed status of the resident-eligible pool. Its statuses are retrospective diagnostics, not prospectively frozen gates for this experiment.

D. Confirmatory and Diagnostic Roles

The learned controller and its hypothesis test were frozen before test-label access. The oracle ceilings, head-

room, interaction contrasts, task-aware oracle, and failure decompositions were computed afterward from the complete sealed matrix in response to the negative result. We label them post-hoc diagnostics throughout. They explain why the confirmatory policy failed; they do not convert that failure into a successful prospective router claim.

IV. Controlled Testbed and Measurement Contract

A. HybridBench as a Diagnostic Testbed

HybridBench is generated deterministically from seed 4622 without an LLM. It contains 800 independent scenario groups, 100 for each of eight task classes. Every group defines a project, current and obsolete documents, an authority conflict, a project–dataset–scheduler chain, personal-memory events, and a tombstone. Answers and evidence identifiers are generated for the experiment. Their opacity minimizes answer recovery from pretrained parametric knowledge, while exact required and forbidden IDs make route failures observable.

The classes are: (i) copy a supplied token without retrieval; (ii) recover a fact from authoritative knowledge; (iii) recover an active memory; (iv) compose knowledge and memory; (v) resolve a temporal update; (vi) resolve an authority conflict; (vii) traverse two knowledge hops; and (viii) abstain when a private fact has been deleted or is missing. The first and last classes require no active evidence citation. The other six collectively form the 90-question evidence-required test stratum.

For a structurally representative temporal-update item, one scenario contains an archived document stating batch size 8 and a newer approved document stating batch size 16 for the same opaque project. The question asks for the current batch size; the strict label requires answer 16 and the newer document ID. Returning 8, citing only the archive, or citing both records fails at least one answer, recall, or precision condition. Other classes reuse the scenario’s knowledge records, memory events, authority levels, links, or tombstone as controlled distractors. This illustration mirrors the generator rules but is not a disclosed test item.

Scenario groups are split 10/50/15/10/15% into pilot, train, calibration, validation, and test: 80, 400, 120, 80, and 120 queries. A scenario appears in one split only; task–split pairs use distinct template families. The leakage audit rejects group overlap, template-family overlap, normalized duplicates, and question-shingle Jaccard similarity at or above 0.95. A task-only train-majority sanity check answered 21 of 90 evidence-required test items (23.33%). This is a construction diagnostic over low-cardinality task vocabularies, not a random-chance or real-world baseline. Test labels remained procedurally sealed until the policy, threshold, and confirmatory analysis plan were frozen; shared repository credentials make this a procedural rather than cryptographic guarantee.

B. Strict Outcome Contract

Each generator must emit a JSON-like object containing an answer, exact evidence IDs, and an abstention flag. Let

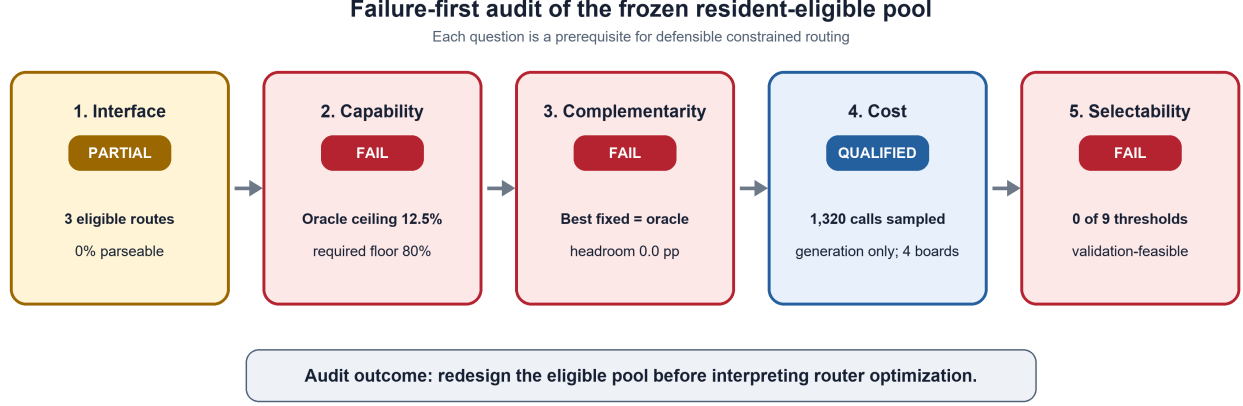


Fig. 1. The five-question audit applied retrospectively to the frozen resident-eligible pool. The statuses summarize existing traces: three eligible Granite-1B actions never parsed; even a per-query oracle reached only 12.5% against the declared 80% requirement; routing headroom was zero; energy telemetry was complete within the declared generation window but spanned physical boards; and no frozen threshold was validation-feasible.

F_{ia} be answer token F1, R_{ia} required-ID citation recall, P_{ia} citation precision, A_{ia} correct abstention, and V_{ia} an indicator for successful parsing, no citation outside the retrieved set, and no forbidden evidence in either retrieved or cited IDs. Strict success is

$$S_{ia} = \mathbb{I} \left[\begin{array}{l} F_{ia} \geq 0.8 \wedge R_{ia} = 1 \wedge P_{ia} = 1 \\ \wedge A_{ia} = 1 \wedge V_{ia} = 1 \end{array} \right]. \quad (4)$$

For evidence-free tasks, recall is one and precision is one only for an empty citation set. A valid deleted/missing response must abstain or explicitly report insufficient evidence without exposing the tombstoned identifiers. The guard is deterministic. It checks syntax and ID membership; it does not prove semantic entailment. We report retrieval completeness separately from citation completeness so that evidence acquisition is not confused with evidence use.

C. Composite Routes

The predeclared catalog contains 22 actions. Pilot rules removed five non-anchor actions, leaving 17 for the clean test matrix. Retrieval variants include BM25 [38], a frozen sentence-bi-encoder derived from dense retrieval practice [39], [40], and bounded hybrid retrieval using reciprocal-rank fusion [41]. Memory variants use flat recency-aware retrieval, hierarchical summary-plus-event retrieval, or graph/temporal traversal. Some actions append a deterministic support guard. These mechanisms are experimental controls, not newly proposed retrieval or memory algorithms.

The matched panel compares each generator directly and under the same composite grounding intervention: hybrid knowledge retrieval, graph/temporal memory, bounded second-hop expansion, a common context budget, exact-citation prompting, and the deterministic guard. Because several components change together, the estimand

is “composite grounding minus direct,” not retrieval in isolation.

The checkpoints are Qwen3-0.6B and Qwen3-4B-Instruct-2507 [42]–[44], Granite-4.0-1B and Granite-4.1-3B [45], [46], and SmolLM3-3B [47]. All revisions, tokenizers, prompt budgets, and FP16 loading settings are immutable. Only the 0.6B and 1B families were available to the online controller; the 3B/4B endpoints are offline references. This eligibility rule was frozen from the resource contract: the 0.6B and 1B checkpoints had to coexist resident on one T4 with at least 15% free VRAM and no CPU or disk offload. Reference checkpoints were instead loaded and unloaded in isolated sequential blocks. Dynamic model loading was outside the route-latency boundary and therefore could not be treated as a zero-cost online action. We call the resulting 11 actions the resident-eligible or deployable pool; this is a specific concurrency-and-latency constraint, not a claim that 3B/4B models are undeployable in general.

D. GPU-Generation-Energy Boundary

We measure generation-window selected-GPU board energy by sampling the selected NVIDIA T4’s NVML board-power reading every 50 ms and applying trapezoidal integration

$$E_{ia} = \sum_{t=1}^{T-1} \frac{P_t + P_{t+1}}{2} (\tau_{t+1} - \tau_t). \quad (5)$$

CUDA is synchronized immediately before and after `model.generate()`. Telemetry records physical GPU UUID, sample count, observed duration, input and output tokens, and failure status. NVML exposes device-level power observations [48]; related GPU-energy instrumentation illustrates why sampling and boundary disclosure matter [49].

TABLE II

Frozen benchmark, retrieval, generation, and pilot-pruning parameters. Candidate depth is evaluated separately for knowledge and active-memory indexes; final context is bounded by the shared token budget.

Component	Frozen setting	Value
Benchmark	Scenario groups / task classes / seed	800 / 8 / 4622
Split	Pilot / train / calibration / validation / test	10 / 50 / 15 / 10 / 15%
Sparse retrieval	BM25 candidate depth; final depth	8; 4
Dense retrieval	all-MiniLM-L6-v2, mean-normalized; candidate/final depth	8 / 4
Hybrid fusion	Equal-list reciprocal-rank fusion constant; final knowledge depth	60; 4
Second hop	Expand dataset aliases found in first-hop evidence; retain final depth	bounded; 4
Memory	Flat dense; hierarchical latest-per-entity; graph linked-recency/fusion	at most 4 events
Prompt budget	Retrieved-context budget / maximum generated tokens	1,536 / 80
Decoding	FP16 model load; greedy decoding	do_sample=false
Resident eligibility	0.6B and 1B coexist on one T4; free VRAM; offload	$\geq 15\%$; none
Pilot retention	Mandatory anchors, or failure $\leq 10\%$, success $\geq 5\%$, energy present	pilot only
Energy	NVML selected-board sampling around model.generate()	50 ms
Uncertainty	Query-cluster bootstrap replicates	10,000

TABLE III

Frozen generator panel and matched direct/grounded endpoints. Only the first two model families satisfied the co-resident online resource contract; references were loaded sequentially. All checkpoints were loaded in FP16 at immutable revisions.

Role	Frozen repository (revision prefix)	Direct	Grounded/verified	Router eligible
Tiny	Qwen/Qwen3-0.6B (c1899de)	A00	M16	yes
Small	ibm-granite/granite-4.0-1b (6a7381b)	A04	A13	yes
Granite reference	ibm-granite/granite-4.1-3b (c065040)	M17	M18	no
Peer reference	HuggingFaceTB/SmolLM3-3B (a07cc9a)	M19	M20	no
Upper reference	Qwen/Qwen3-4B-Instruct-2507 (cdbee75)	M21	A14	no

The reported quantity excludes CPU retrieval and embedding, memory traversal, router computation, parsing, verification, storage, network transfer, host idle power, cooling, and carbon. Route latency is retrieval time plus generator-call duration; policy latency additionally includes its probe and controller components. We make same-board paired claims where possible. Cross-board and route-major comparisons remain descriptive because board, session, thermal, and residency effects were not randomized away.

V. Experimental Protocol

A. Frozen Execution and Provenance

Scientific workers ran in Kaggle sessions with one visible NVIDIA T4; the device mask was set before importing CUDA-aware packages. The initial stage froze source and configuration hashes, route specifications, prompts, model and encoder revisions, decoding, seed 4622, and the Python/Torch/CUDA environment. Later stages rejected a changed experiment identifier, specification hash, source bundle, environment contract, or worker lane. Deterministic decoding used FP16, greedy generation, an 80-token output limit, and a 1,536-token context budget. OOM or context overflow was recorded rather than silently changing precision or budget.

The pilot evaluated 22 actions on 80 queries and pruned five non-anchor actions by the frozen rule. Router fitting used 11 eligible actions over 400 train, 120 calibration, and 80 validation queries, yielding 4,400, 1,320, and 880 route-query traces. The test matrix was not used for fitting,

calibration, thresholding, or fallback selection. The clean release executes each of 17 retained actions on all 120 test queries once, for 2,040 traces.

B. Two-Stage Controller

Stage 0 uses eight query-available features—length, character and digit counts, entity codes, and temporal, conflict, memory, and multi-hop cues—to score direct actions. If no direct action clears the decision constraints, one charged BM25/memory-metadata probe supplies eight retrieval statistics, including top scores, score gaps, counts, and authority values. Stage 1 then scores non-direct actions. If no action qualifies, the controller invokes a validation-selected fail-closed fallback.

For each of five grouped-bootstrap seeds, a histogram gradient-boosting classifier predicts strict success. Isotonic regression calibrates each predictor on the calibration split, following the general requirement that decision confidence be empirically calibrated [50]. Separate quantile regressors predict log energy and latency at $q = 0.90$. The controller uses the minimum calibrated success probability across bootstrap members and mean exponentiated cost predictions. Among routes above threshold τ with predicted latency at most 12 s, it chooses the lowest predicted generation energy. This reject-option design is related to selective prediction [51], but its fallback is not evidence of safety; it is only the declared completion behavior.

C. Frozen Constraints and Amendment

A validation threshold was feasible only if strict success was at least 0.80, execution coverage at least 0.90, and abstention at most 0.20. The 80% value was a predeclared deployment utility requirement, not a prediction of attainable model performance. It was intentionally not relaxed after observing the routes. Every threshold in the original grid $\{0.60, 0.65, \dots, 0.85\}$ produced 8.75% validation success, 100% coverage, and 18.75% abstention, so the validation constraints were infeasible and the original implementation stopped without a policy artifact.

After validation infeasibility was known but before test-label access, amendment stage06-threshold-completion-separation-v1 extended the audit grid to 0.90, 0.95, and 1.00. Outcomes were unchanged. The amendment froze $\tau = 1.0$ to permit completion of the already specified test matrix, recorded `validation_feasible=false`, and prohibited a positive hypothesis claim. The fallback A03 was selected on validation by strict success, then median measured GPU energy, then route identifier. We retain the amendment as a protocol deviation and never describe this fallback as validation-feasible or empirically safe.

D. Confirmatory Estimands

For test query i , let $\pi(i)$ be the frozen policy action and b the fixed A03 fallback baseline. The paired estimands are

$$\Delta_S = \frac{1}{N} \sum_{i=1}^N (S_{i,\pi(i)} - S_{i,b}), \quad (6)$$

$$\Delta_E = \frac{1}{N} \sum_{i=1}^N (E_{i,\pi(i)} - E_{i,b}). \quad (7)$$

The confirmatory claim required all four gates:

$$\begin{aligned} L_{.95}(\Delta_S) &\geq -0.03, & U_{.95}(\Delta_E) &< 0, \\ \text{coverage} &\geq 0.90, & \text{abstention} &\leq 0.20. \end{aligned} \quad (8)$$

If policy and baseline select the same trace, both paired effects are identities at zero. Passing the formal non-inferiority margin in that case does not demonstrate preservation of useful quality; energy reduction still requires a strictly negative upper interval bound.

E. Uncertainty and Diagnostic Analyses

Paired 95% intervals use 10,000 percentile-bootstrap replicates with seed 4622, resampling whole scenario IDs [52]. They quantify variation across the frozen questions, not decoding or repeated-hardware variance. The predeclared analyses cover policy-baseline effects, matched grounded-minus-direct effects, and route-level contrasts.

After the confirmatory failure, we derived the action-pool oracle, routing headroom, unique successes, generator interaction contrasts, task-aware oracle, retrieval-to-utilization decomposition, and answer-F1 threshold sensitivity from the same immutable traces. These are explicitly exploratory and are not multiplicity-adjusted. No model was rerun and no outcome was replaced.

F. Robustness and Hardware Assignment

The diagnostic corruption matrix evaluates A00 direct, A03 hybrid, and A13 grounded/guarded under stale evidence, untrusted conflict injection, missing evidence, and deletion/duplicate ordering, for 1,440 traces. Because the selected policy has zero clean utility, its corruption deltas are interpreted only as a floor effect. The literal compromise marker is a narrow synthetic indicator, not a general injection detector.

Clean routes span four physical T4 boards. Four matched pairs—Qwen3-0.6B, Granite-4.1-3B, SmoLM3-3B, and Qwen3-4B—are same-board within pair. The Granite-4.0-1B pair is cross-board. Tiny BM25, dense, hybrid, and direct endpoints share one board; the tiny memory routes do not. The policy and A03 baseline refer to the same stored trace on every query, so their identity comparison is unaffected by board assignment.

VI. Results

A. RQ1: The Resident-Eligible Pool Has Zero Success Headroom

The post-hoc audit in Table IV separates the opportunity present in the action pool from the learned policy. In the 11-action resident-eligible pool, A00 succeeded on 15 of 120 queries (12.5%). No other eligible route succeeded on any query, so the union of successes was exactly the A00 success set. Therefore $B(\mathcal{A}_{\text{dep}}) = O(\mathcal{A}_{\text{dep}}) = 0.125$ and $H(\mathcal{A}_{\text{dep}}) = 0$. This is an empirical impossibility certificate for strict-success improvement on the sealed matrix: no alternative selector, including a label-aware one, could beat the fixed A00 route with the same actions.

A00 also had the smallest observed mean generation energy in the eligible pool, 54.71 J/query; all ten zero-success alternatives had higher route means. Thus no fixed route improved the observed mean-quality/mean-energy point. A label-aware per-query cost bound is slightly different: while retaining all 15 A00 successes, it chooses A03 for one A00-failure query and lowers the mean by 0.049 J/query (0.089%). The changed trace is same-board. This negligible but nonzero saving is why zero success headroom is not described as a proof of zero cost headroom. Other cross-board comparisons remain descriptive.

The reference actions show that zero headroom was not a property of the testbed. The Granite-4.1-3B direct/grounded pair had 20.8 percentage points of headroom, the SmoLM3 pair 21.7 points, and the Qwen3-4B pair 13.3 points. Across the six offline reference actions, the oracle reached 72.5% versus a 44.2% best fixed route. The complete 17-action matrix reached 73.3%, leaving 29.2 points of headroom. Complementarity existed, but the capable complementary actions were unavailable to the controller. Figure 2 makes this action-set dependence visible: the same test queries have no utility gap under resident eligibility and a substantial gap once the sequential reference endpoints are admitted.

TABLE IV

Post-hoc action-pool routability on the complete sealed-test matrix. The per-query oracle uses outcome labels and is an upper-bound diagnostic, not a deployable policy. Headroom is oracle utility minus the best fixed route.

Pool	Actions	Best route	Best fixed (%)	Oracle (%)	Headroom (pp)
Resident-eligible pool	11	A00	12.5	12.5	0.0
Granite 4.1 3B pair	2	M18	44.2	65.0	20.8
SmolLM3 3B pair	2	M20	34.2	55.8	21.7
Qwen3 4B pair	2	A14	42.5	55.8	13.3
Offline reference pool	6	M18	44.2	72.5	28.3
All retained routes	17	M18	44.2	73.3	29.2

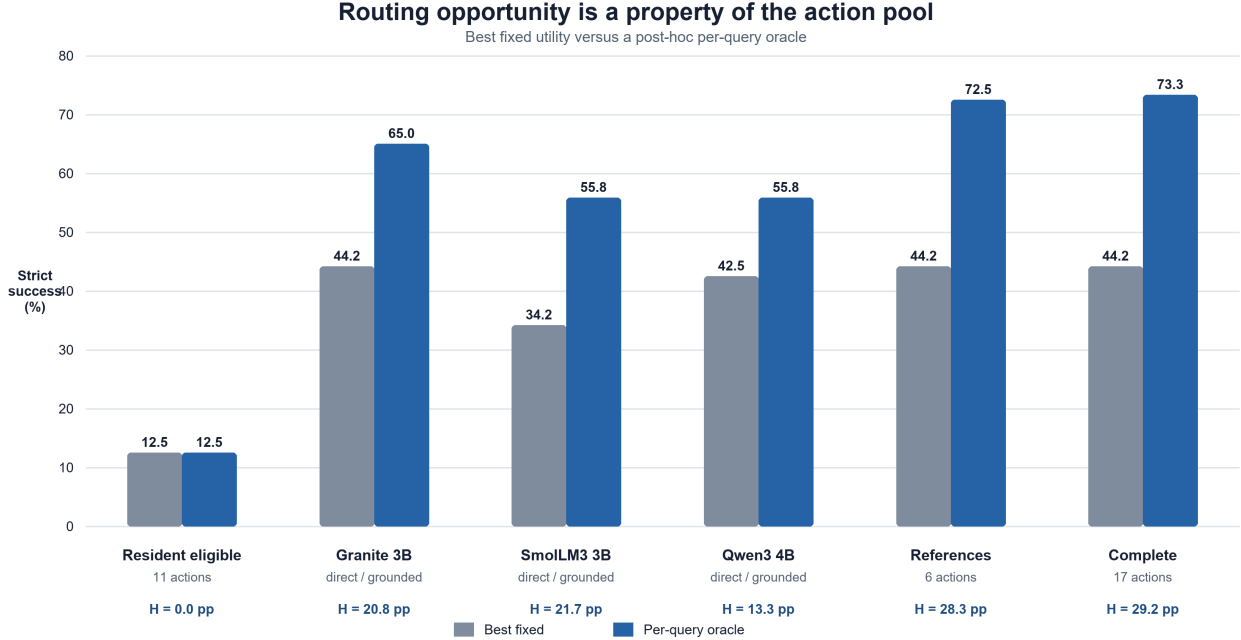


Fig. 2. Best fixed strict success and the post-hoc per-query oracle for six action pools. The resident-eligible pool has no success headroom; each capable direct/grounded reference pair does. Oracle values use labels and quantify an upper bound, not prospective policy performance.

B. RQ2: Retrieval Success Does Not Imply Evidence Use

All five grounded endpoints received identical ordered knowledge and memory IDs on all 120 queries. On the 90 evidence-required items, those lists contained every required ID for 72 queries (80.0%); the two-hop class was complete for 14 of 15. This common acquisition layer provides a controlled basis for comparing what happened after retrieval.

Table V reveals a large retrieval-to-utilization gap. The Qwen3-0.6B grounded endpoint parsed only 32.2% of evidence-task outputs and never produced a complete citation set or a valid answer. The Granite-4.0-1B endpoint parsed none. In contrast, Granite-4.1-3B parsed 95.6%, cited all required evidence on 77.8%, and achieved 58.9% strict success. SmolLM3 parsed every output but cited all required IDs on only 44.4%, yielding 43.3% strict success. Qwen3-4B reached 56.7%. Because retrieved IDs were held constant, differences in those IDs cannot explain the between-generator variation. Formatting, answer formation, citation transfer, and abstention remain observable downstream failure loci; the traces do not provide a

variance decomposition among them.

C. RQ3: Grounding Is a Generator–Task Interaction

Across the equal eight-task mixture, composite grounding changed strict success by -12.5 percentage points for Qwen3-0.6B (95% interval $[-18.3, -6.7]$), $+23.3$ for Granite-4.1-3B ($[10.0, 37.5]$), $+10.8$ for SmolLM3 ($[-2.5, 24.2]$), and $+29.2$ for Qwen3-4B ($[16.7, 41.7]$). The 1B Granite endpoints both scored zero and are not interpreted as a grounding effect. Exploratory difference-in-differences contrasts make the interaction explicit: the Granite-4.1-3B, SmolLM3, and Qwen3-4B effects exceeded the Qwen3-0.6B effect by 35.8 [23.3, 48.3], 23.3 [11.7, 34.2], and 41.7 [29.2, 54.2] points, respectively. Granite-4.1-3B also exceeded SmolLM3 by 12.5 [5.8, 20.0] points. These post-hoc intervals are not multiplicity-adjusted.

Figure 3 shows why an aggregate “RAG helps” statement is misleading. Qwen3-0.6B already had zero success on every evidence-required class; the composite endpoint did not make those answers worse. It lost the 15 direct-copy successes without adding an evidence-task success.

TABLE V

Retrieval-to-utilization decomposition on the 90 evidence-required questions. Every grounded endpoint received identical ordered retrieved IDs; retrieval completeness checks required IDs against those retrieved IDs. Citation completeness is an output property and is therefore reported separately.

Generator	Retrieved (%)	Parse (%)	Cite all (%)	Answer valid (%)	Support valid (%)	Strict (%)
Qwen3 0.6B	80.0	32.2	0.0	0.0	0.0	0.0
Granite 4.0 1B	80.0	0.0	0.0	0.0	0.0	0.0
Granite 4.1 3B	80.0	95.6	77.8	58.9	82.2	58.9
SmolLM3 3B	80.0	100.0	44.4	51.1	67.8	43.3
Qwen3 4B Instruct	80.0	85.6	68.9	56.7	73.3	56.7

Grounding effects reverse across generators and task classes

Grounded minus direct strict success (percentage points; 15 questions per cell)

	Direct copy	Knowledge	Memory	Know. + mem.	Temporal	Authority	Two hop	Deleted
Qwen3 0.6B	-100.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
Granite 4.0 1B	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
Granite 4.1 3B	-66.7	+93.3	+100.0	+20.0	+100.0	+40.0	+0.0	-100.0
SmolLM3 3B	-86.7	+20.0	+100.0	+13.3	+93.3	+33.3	+0.0	-86.7
Qwen3 4B Instruct	-6.7	+93.3	+66.7	+13.3	+100.0	+40.0	+26.7	-100.0

Red: grounding reduced success | Blue: grounding increased success

The 1B Granite row is an integration failure and is not interpreted as model capability.

Fig. 3. Grounded-minus-direct strict success by generator and task class. Each cell contains 15 paired queries. Several composite-grounded endpoints lose direct-task successes, while only the capable references gain strict successes on evidence tasks. The Granite-4.0-1B row is an integration failure.

The capable references display the opposite pattern: direct routes cover evidence-free behavior, whereas composite-grounded routes add knowledge, memory, temporal, and authority successes. Two-hop strict success remains weak despite 93.3% retrieval completeness, further separating evidence acquisition from use.

The same complementarity yields a constructive post-hoc result. A task-aware rule that chooses direct operation for no-retrieval and deleted/missing items and the matched grounded route for the other six classes exactly matches the pair oracle in this matrix. For Granite-4.1-3B, SmolLM3, and Qwen3-4B, it reaches 65.0%, 55.8%, and 55.8% success, improving on always grounding by 20.8, 21.7, and 13.3 points. It also lowers observed generation energy by 14.71, 13.05, and 30.32 J/query, respectively. The rule receives benchmark task labels and is not deployable as-is; it demonstrates a utility pattern that a suitable eligible pool and retrieval-need signal could potentially exploit in a prospective study.

D. Grounded Prompts Increase Measured Generation Work

Every grounded endpoint had a higher mean selected-board generation energy than its direct counterpart. The four same-board increases were 146.72 J/query for Qwen3-0.6B, 119.53 for Granite-4.1-3B, 41.52 for SmolLM3, and 171.71 for Qwen3-4B. The Granite-4.0-1B increase of 3.88 J is cross-board and descriptive. Grounding also added 499–632 mean input tokens and, for the four interpretable pairs, 8–38 output tokens. Per-query energy differences were closely associated with generation-time differences ($r = 0.941$ – 1.000). These are descriptive associations from route-major single runs, not a causal decomposition or an intrinsic model-efficiency ranking.

The apparent tension between higher always-grounded energy and the savings in Table VI is important. The composite grounded action costs more when invoked, but avoiding it on the two evidence-free classes reduces average energy while preserving complementary successes. Conditional invocation, not always-on grounding, creates the useful quality–cost structure.

TABLE VI

Exploratory task-aware choice: direct for no-retrieval and deleted/missing items, grounded otherwise. It uses benchmark task labels and is not a learned router. Effects compare this choice with always grounding; intervals are paired scenario-cluster bootstraps.

Generator	Direct (%)	Grounded (%)	Task-aware (%)	Δ success [95% CI]	Δ GPU J [95% CI]
Qwen3 0.6B	12.5	0.0	12.5	+12.5 [+6.7, +18.3]	-34.97 [-46.12, -24.22]
Granite 4.1 3B	20.8	44.2	65.0	+20.8 [+13.3, +28.3]	-14.71 [-23.07, -7.17]
SmolLM3 3B	23.3	34.2	55.8	+21.7 [+14.2, +29.2]	-13.05 [-18.43, -8.19]
Qwen3 4B Instruct	13.3	42.5	55.8	+13.3 [+7.5, +20.0]	-30.32 [-44.14, -18.18]

TABLE VII

Confirmatory policy result on the sealed test split ($n = 120$). Intervals are 95% scenario-cluster bootstrap intervals.

Quantity	Result
Selected route	A03 for all queries
Strict success	0.0% [0.0, 0.0]
Generation-window GPU energy	143.16 J/query
Energy interval	[138.14, 148.07] J
Policy latency component sum	3.76 s/query
Abstention / execution coverage	15.0% / 100.0%
Policy – baseline success	0.0 pp
Policy – baseline energy	0.00 J/query
Formal margin check	pass; policy = baseline
Energy reduction	fail
Coverage and abstention gates	pass
Joint confirmatory hypothesis	fail

E. RQ4: The Frozen Controller Confirms the Audit Failure

No validation threshold met the frozen success, coverage, and abstention requirements. Under the disclosed completion amendment, the controller selected A03 for all 120 test queries. It achieved 0% strict success, used 143.16 J/query, and had 15% abstention with 100% execution coverage. The fixed baseline was the same A03 trace for every query. Consequently, $\Delta_S = \Delta_E = 0$ by identity, with degenerate intervals. The formal non-inferiority margin passed but is non-informative at zero utility; the energy-reduction gate failed, and the joint confirmatory hypothesis failed.

This outcome is not evidence that A03 was the best action for every query. It is fallback collapse: no predicted action cleared the threshold. The post-hoc oracle adds the stronger diagnosis that even replacing the controller with a perfect strict-success selector could not have improved over A00 within the eligible pool. The learned policy additionally chose the wrong fixed action for the observed test matrix, but action-pool redesign is required before classifier refinement can create routing headroom.

F. Compatibility and Corruption Diagnostics

The retained Granite-4.0-1B actions A04, A12, and A13 never parsed under the frozen prompt/chat-template/runtime combination and repeatedly reached the 80-token limit with a degenerate output. We classify this as a model-prompt-runtime compatibility failure, not evidence that the checkpoint family is generally incapable

of RAG. Repairing the interface after test access would have changed the specification.

Under all four corruptions, the selected policy again used A03 and remained at zero strict success. A zero degradation from a zero clean reference is a floor effect, not robust competence. A00 remained at 12.5% because it ignores evidence, while A13 remained at zero because of the compatibility failure. The synthetic compromise marker was never emitted, but that fact does not establish general prompt-injection resistance.

VII. Discussion: From Route Catalogs to Routable Pools

A. Separate Action-Space Failure from Learning Failure

The central diagnostic is a separation of failure loci. A learned controller can fail because its features, labels, calibration, or decision rule are inadequate. It can also fail because its candidate set does not contain a useful conditional choice. The first is a learning problem; the second is an action-space problem. On the sealed resident-eligible matrix, zero routing headroom proves the latter for strict success. The policy’s universal A03 fallback adds a learning-and-selection failure, but replacing the classifier alone cannot create successes absent from every eligible alternative.

This distinction suggests a routing-readiness order. First validate each model-prompt-runtime interface. Next measure individual route capability. Then compute success unions, unique contributions, and oracle headroom. Only after complementarity is present should one test whether decision-time features predict which route will help. Physical cost comparison and constrained policy optimization follow, not precede, those checks. A route catalog with many names is not necessarily a diverse action pool in outcome space.

B. A Deployment Requirement Is Not an Attainability Claim

The frozen 80% success floor was an operating requirement, whereas the 12.5% oracle ceiling is an empirical property of the eligible test matrix. Conflating the two would make the threshold appear to be a tuning error. The correct inference is that the pool cannot satisfy the declared requirement even with label access. A different application may choose another requirement, but the ceiling remains useful: it shows exactly which requirements are unattainable without changing the action set.

At the same time, a positive oracle gap would not guarantee a successful router. The oracle sees outcomes; a deployment policy sees query and probe features. Future work must demonstrate that complementary successes are predictable out of sample and that the resulting gains survive policy overhead. The audit is a necessary filter, not a replacement for prospective evaluation.

C. Grounding Should Be Conditional on Both Need and Capability

The matched pairs explain why neither “always retrieve” nor “never retrieve” is an adequate general policy. Direct behavior covers the supplied-token class and, for several references, correct abstention on deleted or missing facts. Composite-grounded endpoints add successes in several evidence-required classes for capable generators but lose some evidence-free successes. Because retrieval, memory, prompting, and the deterministic guard change jointly, the average composite-grounded-minus-direct effect changes with workload composition and is not an isolated retrieval effect.

The task-aware oracle makes the design implication concrete. For capable reference generators, conditional grounding improves success and reduces generation energy relative to always grounding. For Qwen3-0.6B, the composite-grounded endpoint never adds an evidence-task success, so recognizing retrieval need alone is insufficient; the system must also know whether the selected generator can use the retrieved contract. This supports generator-conditional evidence policies rather than a single model-agnostic RAG switch.

D. The Interface Is Part of the Action

Three eligible Granite-4.0-1B routes produced no parseable output. Several tiny routes retrieved evidence but serialized citations that the exact-ID guard could not accept. These are operational failures, not cosmetic parser details: an evidence system needs machine-actionable outputs and predictable abstention. However, strict syntax can reject an answer whose intent a human could recover. Reporting parse, answer, citation, support, and strict layers separately prevents that construct choice from being hidden.

A low-cost compatibility screen should therefore precede full route collection. It should test chat-template application, schema adherence, termination, citation serialization, and representative context lengths. If the integration changes, the route specification changes; repairing it after sealed evaluation should create a new version rather than overwrite the trace.

E. Cost Claims Must Follow the Measurement Boundary

The study supports a narrow physical statement: under the frozen same-board pairs, the composite grounded prompt used more selected-T4 energy during generation than the direct prompt. It does not establish total RAG

energy, because retrieval, embedding, memory traversal, routing, verification, and host overhead were excluded. Nor does the strong energy–duration association isolate the cause of the overhead.

The conditional-oracle result also cautions against reporting only per-invocation RAG overhead. An expensive grounded action can lower workload-average energy when it is invoked selectively and replaces wasteful always-on grounding. Credible deployment studies should report the invocation policy, task mixture, useful successes, tokens, latency, and a clearly bounded energy quantity together.

F. Negative Results Need Completion Without Reinterpretation

The Stage-06 amendment permitted artifact completion while forbidding a positive claim. That separation is important. Crashing on validation infeasibility would have hidden the action-space diagnosis; silently relaxing the success requirement would have contaminated the confirmatory test. A fail-closed completion state, immutable traces, and a post-hoc analysis label allow the negative result to remain scientifically useful without turning it into a successful router after the fact.

VIII. Threats to Validity, Reproducibility, and Ethics

A. Construct and External Validity

HybridBench is synthetic and controlled. Its generated identifiers and exact evidence sets support mechanism diagnosis but do not estimate performance on open-domain QA, long-document memory, conversational personalization, or production traffic. The equal eight-class mixture is a design choice, and each test class contains only 15 queries. Model rankings and average grounding effects may change with another task distribution, prompt, context budget, retriever, or verifier. This limitation is especially important given evidence that synthetic context-utilization tests can misrepresent behavior under realistic retrieval [29].

Strict success is an operational conjunction. Exact-ID scoring can reject a recoverable formatting variant, while set membership cannot prove semantic entailment. The layered decomposition mitigates but does not remove this construct limitation. Sensitivity at answer-F1 thresholds 0.5, 0.8, and 1.0 preserves the ranking and effect signs of the capable grounded endpoints (Appendix C), but it does not test alternative citation semantics.

B. Post-Hoc Diagnostics and Statistical Scope

The action-pool oracle, routing headroom, unique-success counts, interaction contrasts, task-aware oracle, and failure layers were specified after the confirmatory policy failed. They use sealed-test outcomes and must not be read as prospective policy estimates. The task-aware oracle is given the benchmark task class, a signal that a real deployment would have to infer. Positive headroom in the reference pool shows that complementary outcomes

exist on this matrix; it does not establish out-of-sample predictability.

Only one deterministic generation was recorded per route–query pair. Bootstrap intervals resample scenarios and do not quantify decoding variance, repeated-run variance, or hardware uncertainty. Exploratory intervals are not corrected for multiple comparisons. The controller is evaluated by replaying a complete frozen route matrix rather than by an independent prospective online deployment.

C. Measurement and Hardware Validity

The energy endpoint covers only the selected GPU board during `model.generate()`. CPU retrieval, dense encoding, memory traversal, routing, parsing, verification, indexing, model loading, uploads, host idle power, network, cooling, and storage are outside the boundary. No idle-power subtraction, whole-system instrumentation, carbon conversion, or token-normalized causal model is reported. Consequently, the work does not measure whole-system energy and does not measure carbon emissions.

Four T4 boards were used, execution was route-major, and routes were not randomized or interleaved. Board, session, temperature, and residency can confound cross-lane or cross-model comparisons. Four matched effects are same-board; the Granite-4.0-1B pair is cross-board. Input and output tokens also differ, so this is not an intrinsic architecture-efficiency leaderboard.

D. Model, Protocol, and Robustness Limits

Different model families use different tokenizers and chat templates. The Granite-4.0-1B integration failure is specific to the frozen model–prompt–runtime triple. It cannot support a general capability claim about the checkpoint. Five non-anchor actions were pilot-pruned. The 3B/4B references were excluded because they were run sequentially, whereas eligible models had to coexist on one T4 with 15% free VRAM and no offload; model-loading time was not in the route-latency boundary. This resident-set constraint is defensible for the frozen controller but not universal. Dynamic loading, another accelerator, or a different eligible set would induce another routing problem.

The threshold-completion amendment occurred after validation infeasibility was known but before test labels were accessed. It is disclosed, preserved in the policy artifact, and prohibits a positive router claim, but it remains a protocol deviation. The test seal was procedural because collaborators shared repository access. The robustness matrix covers only three anchors and four synthetic corruptions; its all-zero selected-policy outcome cannot demonstrate a defense.

E. Reproducibility and Ethics

The release contains 2,040 clean and 1,440 robustness traces with complete declared generation-window telemetry, frozen specifications, checksums, route cards, analysis

code, and the amended policy. Artifacts were consolidated without deleting worker branches, restored into a fresh directory, and hash-verified before release. This supports audit and exact re-analysis, although reproducibility on a different T4 may not reproduce the same power readings.

The generated benchmark contains no personal information; memory records and deletion events are synthetic. No external sponsor influenced the study. Because the work neither measures whole-system energy nor converts electricity to emissions, it makes no claim about avoided carbon or environmental impact. If the dataset repository is private during review, an anonymous checksum-preserving archive is required for reviewers.

IX. Conclusion

This study asks a question that precedes router optimization: is the eligible RAG action pool routable? The best-fixed/oracle decomposition gives a direct answer. For the 11 resident-eligible actions, both utilities were 12.5%; all successes came from one direct route, so strict-success headroom was zero. The learned controller’s validation infeasibility and universal zero-success fallback were therefore not only a poor policy outcome. They exposed an action set that no selector could use to improve strict success on the sealed matrix.

The full experiment also identifies a pool with nonzero utility complementarity. Capable 3B/4B direct and grounded pairs had 13.3–21.7 percentage points of oracle headroom. With identical retrieved evidence, their post-retrieval success differed sharply across generators and tasks. A task-aware choice between direct and grounded operation improved both strict success and measured generation energy relative to always grounding. The observed opportunity lay outside the controller’s resident-eligible pool and was not tested with a prospective selector.

The resulting recommendation is to establish interface validity, capability, complementarity, cost comparability, and validation selectability before attributing value to an energy-aware RAG router. Routing algorithms can exploit differences; they cannot manufacture differences that the action pool does not contain.

Appendix A

Complete Route Catalog and Hardware Assignment

Table VIII reports every predeclared route and identifies the 17 retained clean-test routes. “Retained” does not mean “best”: all matched model anchors were mandatory even when pilot quality was weak. The route ID prefix is a stable experimental identifier, not a rank.

Table IX anonymizes the physical board UUIDs while preserving comparability structure. Routes sharing an alias used the same T4. Separate robustness conditions were executed on separate boards and are not included in cross-condition energy comparisons.

TABLE VIII

Predeclared route catalog and pilot retention. A05–A07, A11, and A15 were removed only by the frozen pilot rule; all mandatory matched endpoints were retained.

ID	Generator	Mechanism	Test retained	Success	GPU J
A00	Qwen3 0.6B	Direct	yes	12.5%	54.71
A01	Qwen3 0.6B	BM25 knowledge	yes	0.0%	157.68
A02	Qwen3 0.6B	Dense knowledge	yes	0.0%	150.41
A03	Qwen3 0.6B	Bounded hybrid knowledge	yes	0.0%	143.16
A04	Granite 4.0 1B	Direct	yes	0.0%	256.98
A05	Granite 4.0 1B	BM25 knowledge	no	–	–
A06	Granite 4.0 1B	Dense knowledge	no	–	–
A07	Granite 4.0 1B	Bounded hybrid knowledge	no	–	–
A08	Qwen3 0.6B	Flat memory	yes	0.0%	148.96
A09	Qwen3 0.6B	Hierarchical memory	yes	0.0%	170.32
A10	Qwen3 0.6B	Graph/temporal memory	yes	0.0%	171.49
A11	Granite 4.0 1B	Graph/temporal memory	no	–	–
A12	Granite 4.0 1B	Hybrid knowledge + flat memory	yes	0.0%	258.54
A13	Granite 4.0 1B	Hybrid + graph memory + guard	yes	0.0%	260.85
A14	Qwen3 4B Instruct	Hybrid + graph memory + guard	yes	42.5%	256.13
A15	Granite 4.0 1B	BM25 + hierarchical memory + guard	no	–	–
M16	Qwen3 0.6B	Matched grounded/verified	yes	0.0%	201.44
M17	Granite 4.1 3B	Matched direct	yes	20.8%	57.14
M18	Granite 4.1 3B	Matched grounded/verified	yes	44.2%	176.68
M19	SmolLM3 3B	Matched direct	yes	23.3%	107.42
M20	SmolLM3 3B	Matched grounded/verified	yes	34.2%	148.93
M21	Qwen3 4B Instruct	Matched direct	yes	13.3%	84.42

TABLE IX

Anonymized physical-T4 assignment for clean-test routes. Within-board contrasts avoid board identity as a direct confound; cross-board energy contrasts remain descriptive.

Route	Physical board alias
A00_tiny_direct	GPU-D
A01_tiny_bm25	GPU-D
A02_tiny_dense	GPU-D
A03_tiny_hybrid	GPU-D
A04_small_direct	GPU-C
A08_tiny_memory_flat	GPU-A
A09_tiny_memory_hier	GPU-A
A10_tiny_memory_graph	GPU-A
A12_small_hybrid_both	GPU-B
A13_small_hybrid_verified	GPU-B
A14_upper_hybrid_verified	GPU-B
M16_tiny_grounded_verified	GPU-D
M17_granite_direct	GPU-D
M18_granite_grounded_verified	GPU-D
M19_peer_direct	GPU-A
M20_peer_grounded_verified	GPU-A
M21_upper_direct	GPU-B

TABLE X

Selected clean-test route results. GPU J denotes mean generation-window selected-board energy; the complete matrix and hardware assignment are in Appendix A.

ID	Success (%)	GPU J	Median s
M18	44.2	176.68	2.65
A14	42.5	256.13	3.79
M20	34.2	148.93	2.34
M19	23.3	107.42	1.67
M17	20.8	57.14	0.45
A00	12.5	54.71	1.44
A03	0.0	143.16	3.47

Appendix B

Scoring and Router Contract

A. Token and Citation Metrics

Let $n_t^{\hat{y}}$ and n_t^y denote predicted and target token counts. Answer precision, recall, and F1 are

$$P_{\text{ans}} = \frac{\sum_t \min(n_t^{\hat{y}}, n_t^y)}{|T_y|}, \quad (9)$$

$$R_{\text{ans}} = \frac{\sum_t \min(n_t^{\hat{y}}, n_t^y)}{|T_y|}, \quad (10)$$

$$F_1 = \frac{2P_{\text{ans}}R_{\text{ans}}}{P_{\text{ans}} + R_{\text{ans}}}. \quad (11)$$

For required IDs R_q and cited IDs C_{qr} ,

$$R_{\text{cite}} = \frac{|C_{qr} \cap R_q|}{|R_q|}, \quad P_{\text{cite}} = \frac{|C_{qr} \cap R_q|}{|C_{qr}|}. \quad (12)$$

Recall is one when no evidence is required. Precision for an empty citation set is one only when no evidence is required. Strict success additionally requires $C_{qr} \subseteq G_{qr}$ for retrieved IDs G_{qr} and $F_q \cap (C_{qr} \cup G_{qr}) = \emptyset$ for forbidden IDs F_q .

B. Router Features and Hyperparameters

The eight query-only features are token count, character count, digit count, entity-code count, temporal-term count, conflict-term count, memory-term count, and multi-hop-term count. The eight probe features are document top score, document score gap, document count, memory top score, memory score gap, memory count, maximum authority, and authority range.

Each bootstrap model uses histogram gradient boosting with learning rate 0.06, 180 iterations, at most 15 leaves, and $L_2 = 1$. Success is calibrated by isotonic regression; energy and latency use 0.90-quantile loss on log outcomes. Bootstrap seeds are 4622, 1701, 31415, 27182, and 65537.

C. Validation Amendment Record

The frozen policy records the following non-negotiable fields:

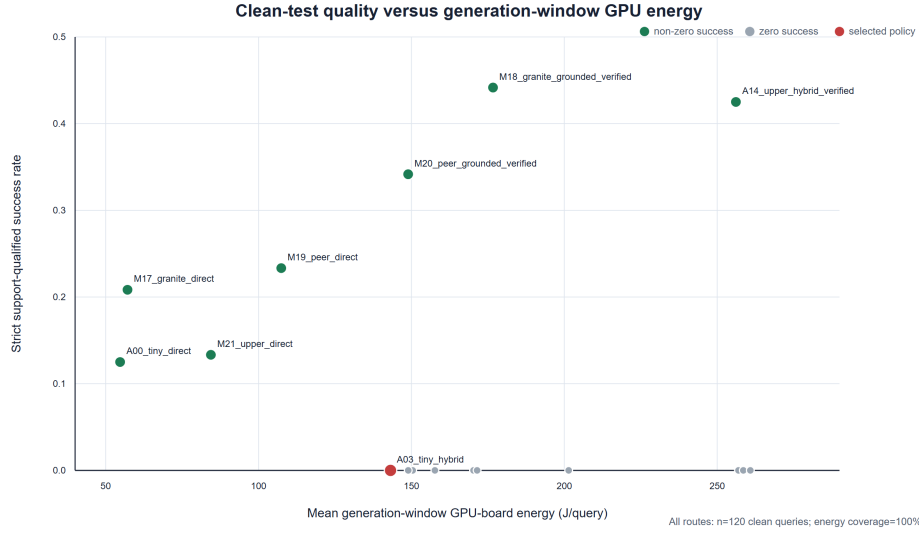


Fig. 4. Strict success versus mean generation-window selected-board energy for all 17 retained clean routes. Model size, mechanism, and board assignment vary, so this is a descriptive map rather than a causal energy frontier.

- amendment ID: stage06-threshold-completion-separation-v1;
- timing: after validation, before test access;
- validation feasible: false;
- selected threshold: $\tau = 1.0$;
- selection mode: fail-closed-tau-1.0-validation-infeasible;
- positive hypothesis claim permitted: false.

Every tested threshold from 0.60 to 1.00 had 8.75% validation success, 100% execution coverage, and 18.75% abstention. The frozen policy file has SHA-256 fda2bda0a715227162ceead0ff032f2228999f6dc9f4bb22890ec23a4acdab14 in the Stage-06 manifest.

Appendix C

Scenario-Specific Grounding Effects

Table XI expands the aggregate matched effects into the eight 15-query test strata. It shows why an always-grounded average must not be interpreted as the conditional effect of retrieval when retrieval is required.

TABLE XI

Direct $\rightarrow$ grounded strict success by task class (%). NR: no retrieval; KO: knowledge only; MO: memory only; KM: knowledge plus memory; TU: temporal update; AC: authority conflict; MH: multi-hop; DM: deleted/missing.

Generator	NR	KO	MO	KM	TU	AC	MH	DM
Qwen3 0.6B	100 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0
Granite 4.0 1B	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0
Granite 4.1 3B	66.7 $\rightarrow$ 0	0 $\rightarrow$ 93.3	0 $\rightarrow$ 100	0 $\rightarrow$ 20.0	0 $\rightarrow$ 100	0 $\rightarrow$ 40.0	0 $\rightarrow$ 0	100 $\rightarrow$ 0
SmolLM3 3B	86.7 $\rightarrow$ 0	0 $\rightarrow$ 20.0	0 $\rightarrow$ 100	0 $\rightarrow$ 13.3	0 $\rightarrow$ 93.3	0 $\rightarrow$ 33.3	0 $\rightarrow$ 0	100 $\rightarrow$ 13.3
Qwen3 4B	6.7 $\rightarrow$ 0	0 $\rightarrow$ 93.3	0 $\rightarrow$ 66.7	0 $\rightarrow$ 13.3	0 $\rightarrow$ 100	0 $\rightarrow$ 40.0	0 $\rightarrow$ 26.7	100 $\rightarrow$ 0

TABLE XII

Matched direct-to-grounded transfer on the same 120 sealed-test questions. Success effects are percentage points; energy effects are generation-window selected-GPU board joules per query. “Same board” indicates whether both endpoints ran on the same physical T4.

Generator	Direct (%)	Grounded (%)	Δ success [95% CI]	Δ GPU J [95% CI]	Same board
Qwen3 0.6B	12.5	0.0	-12.5 [-18.3, -6.7]	+146.72 [+141.56, +151.62]	yes
Granite 4.0 1B	0.0	0.0	+0.0 [+0.0, +0.0]	+3.88 [+1.73, +6.14]	no
Granite 4.1 3B	20.8	44.2	+23.3 [+10.0, +37.5]	+119.53 [+108.73, +129.74]	yes
SmolLM3 3B	23.3	34.2	+10.8 [-2.5, +24.2]	+41.52 [+35.47, +47.50]	yes
Qwen3 4B Instruct	13.3	42.5	+29.2 [+16.7, +41.7]	+171.71 [+157.74, +185.24]	yes

TABLE XIII

Descriptive same-board grounding overhead. Deltas are grounded minus direct; $r(\Delta J, \Delta t)$ is the per-query Pearson association between energy and generation-time differences. Route-major, single-run execution precludes a causal decomposition.

Generator	Δ input tok.	Δ output tok.	Δ generation s	Δ GPU J	$r(\Delta J, \Delta t)$
Qwen3 0.6B	+631.6	+37.5	+1.669	+146.72	0.941
Granite 4.1 3B	+499.7	+28.1	+1.774	+119.53	0.999
SmolLM3 3B	+499.4	+8.4	+0.630	+41.52	0.996
Qwen3 4B Instruct	+631.6	+33.0	+2.592	+171.71	1.000

TABLE XIV

Independently scored outcome components on all 120 clean questions for selected routes; the columns are not a monotone funnel. “Cite all” is exact required-ID recall and is automatically one on the 30 evidence-free items, creating a 25% floor. “Support” also requires parseability, exact citation precision, no invalid citation, and no forbidden exposure.

Route	Parse (%)	Answer valid (%)	Cite all (%)	Support (%)	Abstention correct (%)	Strict (%)
A00	100.0	16.7	25.0	19.2	65.0	12.5
A03	46.7	15.8	25.0	0.0	72.5	0.0
A13	0.0	12.5	25.0	0.0	12.5	0.0
M16	35.0	12.5	25.0	10.8	14.2	0.0
M18	96.7	44.2	83.3	74.2	70.0	44.2
M20	100.0	48.3	58.3	55.8	68.3	34.2
A14	84.2	47.5	76.7	67.5	62.5	42.5

TABLE XV

Strict-success sensitivity to the answer-F1 threshold for capable grounded endpoints. All other evidence, abstention, and parsing requirements remain unchanged.

Generator	$F_1 \geq 0.5$	$F_1 \geq 0.8$	$F_1 = 1.0$
Granite 4.1 3B	55.8	44.2	41.7
SmolLM3 3B	35.0	34.2	32.5
Qwen3 4B Instruct	46.7	42.5	40.8

TABLE XVI

Selected-policy outcomes under four evidence corruptions. Zero clean and corrupted success makes every success delta a floor effect.

Condition	Success (%)	Abstain (%)	Coverage (%)
conflict injection	0.0	15.8	100.0
deletion duplicates	0.0	15.0	100.0
missing	0.0	18.3	100.0
stale	0.0	15.0	100.0

Appendix D

Reproducibility and Artifact Manifest

A. Frozen Counts and Identifiers

The release identifier is `e2am-memrag-v3r1`; the execution-spec SHA-256 is `1c8f29fd250b87d3546c6ff3d128dc3fe6600bc798c3c0b0b8c98e0b95c76cfc`. The visible Hugging Face release commit is `0b2405d9cca43fd04e35f792fdc4664405154fc6`; the paper-release branch commit is `00fa353f273f3a4b3d57a0b998301c85a1bc098b`. Eleven Stage-09 artifacts totaling 11,528,142 bytes passed byte and SHA-256 verification.

The benchmark contains 5,600 documents and 2,400 memory events across 800 scenario groups. After tombstone processing, one selected-seed memory per scenario remains active. The clean release contains 17 routes, 120

query IDs, and 2,040 rows; the robustness release contains three routes, four conditions, 120 query IDs, and 1,440 rows.

B. Rebuild Procedure

The repository script `scripts/build_manuscript_assets.py` reads the checksum-verified release, reconstructs the frozen deterministic test metadata, and regenerates manuscript CSVs, tables, and vector figures. This includes pool oracles, headroom, task-aware choices, failure layers, interaction contrasts, and threshold sensitivity; it never invokes a model. `scripts/validate_manuscript.py` checks numeric claims, bibliography keys, references, measurement-boundary language, artifact existence, and protocol-amendment disclosure. The complete source is compiled with Tectonic 0.16.9 using `paper/manuscript/main.tex`; the same sources are compatible with Overleaf/pdfLaTeX.

C. Availability

Code and paper sources are hosted at <https://github.com/Shanmuk4622/E2AM-MemRAG>. Frozen traces reside in the Hugging Face dataset `Shanmuk4622/E2AM-MemRAG-Traces`. Repository visibility must be checked at submission time. If the trace repository is not public, the authors should provide a checksum-identical anonymous archive or grant reviewer access rather than claiming unrestricted availability.

D. Author Contribution and Conflicts

Shanmukesh Bonala conceived the study, implemented the benchmark and pipeline, executed and audited the experiments, analyzed the data, and wrote the manuscript. The author declares no competing interests and no external funding for this study.

TABLE XVII

Released trace integrity and outcome audit. Both sets had zero duplicate unit IDs, zero non-finite audited values, zero divergent route hashes, and zero execution failures.

Set	Rows	Routes	Strict successes	Parse	Support	Energy coverage
clean	2040	17	229 (11.2%)	58.4%	18.0%	100.0%
robustness	1440	3	60 (4.2%)	49.8%	6.7%	100.0%

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